

## CHAPTER 9

# Forward-scatter from Meteor Trails

One of the most intriguing scientific aspects of oblique scattering from meteor trails is that, by placing transmitter and receiver several hundred kilometers apart, one may observe considerably fainter and higher meteors than when the same equipment is concentrated at one spot on the earth's surface. True, more sophisticated methods are required to take full advantage of this very desirable raising of the underdense trail ceiling. The ellipsoidal geometry of forward-scatter introduces ambiguities in the measurement of some of the parameters of the meteor, which can be overcome only by deploying additional observing equipment over the countryside. Despite the complexities this challenge to the experimenter's ingenuity is not going unanswered. Methods of measuring spatial position, trail orientation, and meteor velocity are now being developed. If successful, they should greatly expand our present observational limits. Fortunately, these particular difficulties are no handicap in the practical application of forward-scatter to radio communications, although some other problems do come up. Engineered systems for point-to-point communication via meteor trails are now being put into commercial and military service in several countries. Some examples of these equipments will be examined later in this chapter.

We shall first develop the modifications and extensions of the backscatter theory which are needed to cover the forward-scatter situation. In the previous chapter we might have worked out a generalized theory, from which the backscatter and forward-scatter results could be deduced as special cases. The degree of appreciation of such a broad attack would probably vary greatly with the mathematical background of the reader. The particular plan we have elected to follow, in this as in the earlier chapters, has been hopefully designed to take him painlessly through some of the more prickly mathematical thickets without unduly obscuring his view of the forest of physical evidence.

**9-1. Forward-scatter from the underdense trail.** We will not need to rework the theory of Chap. 8 from first principles, as an examination of the forward-scatter geometry suggests that only three relatively simple

mechanical modifications are required to adapt the backscatter theory to the oblique-incidence case.

The first adjustment is to reinsert the  $\sin^2 \gamma$  term in the pattern of the scattering cross section of the trail electron. In three dimensions the amplitude pattern of the electron reradiating as a Hertzian dipole is a circle rotated about a tangent—a tight toroid, or a holeless doughnut. There is no scattered field in either direction along the dipole axis, this axis being coincident with the incident electric vector. The expression for the power received from a single electron given in Eq. (8-1) will therefore read

$$\Delta P_R = \frac{P_T G_T G_R \sigma_e \sin^2 \gamma}{64\pi^2 R_1^2 R_2^2} \quad (9-1)$$

where  $R_1$  is the range from the transmitter to the trail and  $R_2$  the distance from the trail to the receiver. The transmitting and receiving antenna gains may not be identical and so are separated in Eq. (9-1).

The major lobe of the backscatter reradiation pattern of a meteor trail lies in the plane which contains the line of sight and which is also normal to the trail axis. This pattern will be a circle if the electric vector of the incident radiation is parallel to the trail axis, or it will be a figure of eight if the electric vector is perpendicular to the trail. The general condition for forward scattering is that the angle between the incident ray and the trail should equal the angle between the reflected ray and the trail. Reflected rays satisfying this condition will lie on a cone with the same axis as the trail. The dipole-pattern factor will attenuate the power level in certain directions along the cone, depending on the trail orientation, incident polarization, and the spatial geometry. The use of circularly polarized waves would smooth out this effect, though at the expense of an average power loss of 3 db over the power attainable with the optimum linear polarization. ★

It will be noted from Eq. (8-6) that the equivalent backscatter echoing area of an underdense trail is  $\sigma_e q^2$  multiplied by  $R_0 \lambda / 2$ . The last quantity is the square of the length of half the first Fresnel zone. In our second modification to the backscatter theory we shall assume that the new equivalent echoing area will be  $\sigma_e q^2 \sin^2 \gamma$  times the square of half the first Fresnel zone for forward-scatter. This half length  $f$  must be found in terms of wavelength and the geometrical parameters. The center of the first Fresnel zone (the  $t_0$  point) will be at the point of tangency of the trail and an ellipsoid of revolution (i.e., a prolate spheroid) whose foci are at the transmitter and receiver. Figure 9-1 outlines the pertinent geometry. The transmitter  $T$ , receiver  $R$ , and the point of tangency are in one plane called the *plane of propagation*. The meteor trail lies in the tangent plane which is at right angles to the propagation plane. It is a property

of the prolate spheroid that the bisector of the angle  $2\phi$  between  $R_1$  and  $R_2$  is normal to the tangent plane. The angle  $2\phi$  is termed the forward-scatter angle. Let the meteor trail assume any arbitrary orientation in the tangent plane, defined by  $\beta$ , the angle between the trail and the plane of propagation. Next, draw the paths  $R'_1$  and  $R'_2$  to one end of the first Fresnel zone, indicated by the segment  $f$  on the diagram. The angle  $\omega$  between  $f$  and  $R_1$  is the supplement of the angle between  $f$  and  $R_2$ , as can be seen by reflecting the rays  $R_2$  and  $R'_2$  in the tangent plane. Note also that  $\cos \omega = \sin \phi \cos \beta$ .

In the triangle with sides  $R_1$ ,  $R'_1$ ,  $f$ , we have

$$(R'_1)^2 = R_1^2 + f^2 - 2R_1f \cos \omega = (R_1 - f \cos \omega)^2 + f^2 \sin^2 \omega$$

Similarly,  $(R'_2)^2 = (R_2 + f \cos \omega)^2 + f^2 \sin^2 \omega$ . Since  $f^2 \sin^2 \omega$  is small by comparison with  $(R_1 - f \cos \omega)^2$  or  $(R_2 + f \cos \omega)^2$  we may approxi-

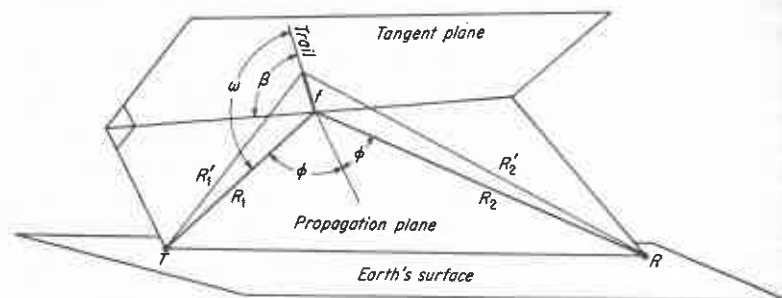


FIG. 9-1. The geometry of forward-scatter involved in the calculation of  $f$ , the length of one-half of the first Fresnel zone.

mate to the square roots as follows:

$$R'_1 + R'_2 = R_1 + R_2 + \frac{f^2 \sin^2 \omega}{2} \left( \frac{1}{R_1 - f \cos \omega} + \frac{1}{R_2 + f \cos \omega} \right)$$

Only at this stage are we able to neglect  $f \cos \omega$  in comparison with either  $R_1$  or  $R_2$ .

The condition for  $f$  to be one-half of the first Fresnel zone is

$$R'_1 + R'_2 = R_1 + R_2 + \frac{\lambda}{2}$$

It follows directly that

$$f = \left[ \frac{\lambda R_1 R_2}{(R_1 + R_2)(1 - \sin^2 \phi \cos^2 \beta)} \right]^{1/2} \quad (9-2)$$

The final modification to the earlier theory arises when we are summing up the phase differences between the waves scattered from electrons in

different parts of the trail, as in Sec. 8-5. In place of the single datum plane of zero phase (Fig. 8-5) we now have two planes. The datum plane normal to the incident ray is shown as  $MM'$  in Fig. 9-2 and the plane for the emergent ray as  $NN'$ . The net phase angle (for the ray drawn) is

$$\begin{aligned} \frac{2\pi}{\lambda} (a + b) &= \frac{2\pi}{\lambda} r \left[ \sin \left( \frac{\pi}{2} - \phi - \theta \right) + \sin \left( \frac{\pi}{2} - \phi + \theta \right) \right] \\ &= \frac{4\pi}{\lambda} r \cos \theta \cos \phi \end{aligned}$$

The right-hand side is a maximum at  $\theta = 0$  or  $\pi$  and is zero at  $\theta = \pi/2$  or  $3\pi/2$ . For comparison, the dashed line on Fig. 9-2 shows the length of the backscatter path, referred to the plane  $PP'$ , for which the phase angle

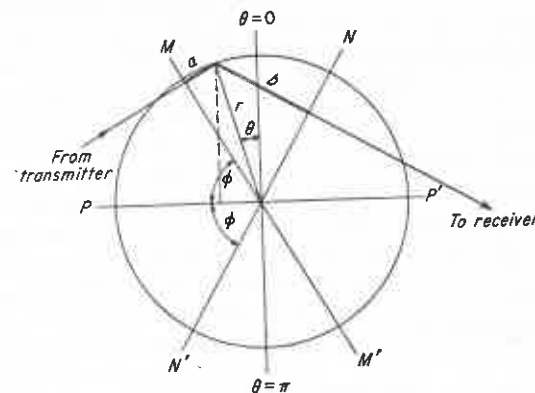


FIG. 9-2. The geometry for the calculation of the phase of the wave scattered from a trail electron.

is  $(4\pi/\lambda)r \cos \theta$ . This shows that we need only to include the factor  $\cos \phi$  to convert the backscatter analysis to the forward-scatter case. In effect,  $\lambda$  has now become  $\lambda \sec \phi$ . Incidentally, to simplify the diagram, Fig. 9-2 was drawn with the trail perpendicular to the plane of propagation, but the conclusion is general and applies to any trail orientation.

To sum up, the power delivered to the receiver after the meteor trail is formed is

$$\begin{aligned} P_R &= \frac{P_T G_T G_R \lambda^3 \sigma_e}{64\pi^3} \frac{q^2 \sin^2 \gamma}{(R_1 R_2)(R_1 + R_2)(1 - \sin^2 \phi \cos^2 \beta)} \\ &= 5 \times 10^{-22} \frac{P_T G_T G_R \lambda^3 q^2 \sin^2 \gamma}{(R_1 R_2)(R_1 + R_2)(1 - \sin^2 \phi \cos^2 \beta)} \text{ watts} \quad (9-3) \end{aligned}$$

This reduces to Eq. (8-7) for the backscatter case, when  $R_1 = R_2$ ,  $\phi = 0^\circ$ ,  $\beta = 90^\circ$ , and  $\gamma = 90^\circ$ .

Substitution of  $\lambda \sec \phi$  for  $\lambda$  in Eq. (8-23) leads to

$$\frac{P_R(t)}{P_R(0)} = e^{-(32\pi^2 D t / \lambda^2 \sec^2 \phi)} e^{-(8\pi^2 r_0^2 / \lambda^2 \sec^2 \phi)} \quad (9-4)$$

and the power attenuation of the echo (for  $R_1 = R_2 = 150$  km) is

$$\text{Loss in db} \simeq 3.8 \times 10^5 \frac{D}{(\lambda \sec \phi)^{3/2} V} + 3.4 \times 10^2 \left( \frac{r_0}{\lambda \sec \phi} \right)^2 \quad (9-5)$$

Equation (8-24) for the echo decay time becomes

$$T_{un} = \frac{\lambda^2 \sec^2 \phi}{16\pi^2 D} \quad (9-6)$$

Some very interesting conclusions may be drawn from these relations. The expression for  $f$  in Eq. (9-2) shows that the size of the first (or any other) Fresnel zone depends on four parameters,  $R_1$ ,  $R_2$ ,  $\phi$ , and  $\beta$ , all of which have to be determined before one can entertain the thought of measuring velocities by forward-scatter. The diffraction-echo fluctuations are indeed present, just as in the backscatter case, and it is not difficult to produce the analytical formulas describing them. Of more immediate importance is the fact that substituting  $\lambda \sec \phi$  for  $\lambda$  increases the decay time by  $\sec^2 \phi$ . Simultaneous observations of echoes by forward-scatter and backscatter have confirmed this within the limits of experimental error (Villard, Peterson, Manning, and Eshleman, 1953, 1956; McKinley and McNamara, 1956). Since  $\sec^2 \phi$  may be as great as 25 or more for stations spaced at 1,000 to 1,500 km, the duration enhancement can be very significant, particularly for the communications application.

When  $\beta = 90^\circ$  the forward-scatter echo power from a given meteor will at least be equal to that obtained from a backscatter radar at a range  $R = [(R_1 R_2)(R_1 + R_2)/2]^{1/2}$ . When  $\beta = 0^\circ$  the lengths of the Fresnel zones increase as  $\sec \phi$ , with a corresponding increase in echo power of  $\sec^2 \phi$ . This power increase tends to be counteracted by the attenuation due to the increase in the ranges, which goes hand in hand with the increase in  $\phi$ .

The meteor scientist will appreciate the very significant increase in the height of the underdense trail ceiling which is implied by Eq. (9-5). To see what this means quantitatively let us choose  $\sec \phi = 3$  as a very conservative estimate of the average value of  $\sec \phi$  for typical forward-scatter systems. If we go back to Fig. 8-7 and multiply the wavelength scale by three, we see that the ceiling is raised by about 7 km. According to Fig. 8-8, the 40-db cutoff for  $\lambda = 10$  m and  $V = 75$  km/sec will be shifted from  $q \simeq 10^{14}$  to  $q \simeq 3 \times 10^{12}$ . The theoretical limit of detectability for meteors of a given velocity is therefore increased by about four

magnitudes. The system sensitivity must also be sufficient to detect the smaller meteors, of course, but the 40-db increase required to extend the sensitivity of our typical 10-m radar (on which the curves of Fig. 8-11 are based) from the tenth to the fifteenth magnitude is well within practical engineering capabilities. In fact, the Stanford group (Gallagher, 1958) and the National Bureau of Standards (Bowles, 1958) have already gone beyond this.

The meteor astronomer still awaits the development of suitable forward-scatter instrumentation for the determination of individual radiants and velocities of these very faint meteors. In the meantime he can carry out statistical studies of shower and nonshower rates, though the interpretation of these observations in terms of hypothetical distributions of meteors over the celestial sphere is not made easy by the unavoidable ambiguities of the simple techniques (Hines, 1955, 1958; Forsyth, Hines, and Vogan, 1955; Bain, 1960). There are so many unknown parameters that the data can be adjusted to fit a wide variety of possible distributions—which would amuse the shades of Schiaparelli and von Niessl! The meteor-communications engineer can hardly be expected to delay his work until a comprehensive description of the orbital distributions of faint meteors becomes available, however useful this would be in predicting the performance of his links. (It may even turn out that the very faint meteor distributions do not repeat predictably.) For him the proof of the pudding lies not in the mystery of its ingredients but in the nourishment it imparts. He can set up his radio circuits and accumulate quite easily all the propagational information that is both necessary and sufficient for the development of his systems.

**9-2. Forward-scatter from the overdense trail.** A satisfactory theory of the echoing properties of the overdense trail has not been worked out for the general forward-scatter case, although the formulas of the backscatter model—itsself not very well determined in several respects—have been converted to the oblique-incidence situation through a judicious admixture of analogy, analysis, and intuition (Hines and Forsyth, 1957; Forsyth, 1958). Manning (1959a) has used geometrical ray-tracing theory to investigate in a semiquantitative manner the effects of refraction outside the critical density radius. Attempts by Keitel (1955) to obtain exact solutions through the application of the wave theory of physical optics were limited to a few special cases. Kaiser and Closs (1952) and Herlofson (1951) laid some of the groundwork for this difficult task, and their approximate solutions agreed reasonably well with the precise calculations of Keitel's electronic computer. Eshleman's early work in this field has been used freely by a number of investigators, though he did not get around to publishing much of it in referable form until six years later (Eshleman, 1955).

The heuristic conversion of the backscatter equations to the forward-scatter case is essentially accomplished through the  $f$  and  $\lambda \sec \phi$  factors of the previous section. The maximum equivalent echoing area of the critical-radius cylinder will now be  $2f^2(qr_e/\epsilon)^{1/2}$  by analogy with the development in Sec. 8-10. Thus we may write

$$P_R = \frac{P_T G_T G_R \lambda^3 r_e^{1/2}}{32\pi^3 \epsilon^{1/2}} \frac{q^{1/2} \sin^2 \gamma}{(R_1 R_2)(R_1 + R_2)(1 - \sin^2 \phi \cos^2 \beta)}$$

$$= 3.2 \times 10^{-11} \frac{P_T G_T G_R \lambda^3 \sin^2 \gamma}{(R_1 R_2)(R_1 + R_2)(1 - \sin^2 \phi \cos^2 \beta)} q^{1/2} \quad (9-7)$$

In practice, for the overdense trail it is found that the signal power falls off less rapidly than  $\sin^2 \gamma$  as  $\gamma$  departs from  $90^\circ$ . This is partly because of the mechanics of reflection at the virtually metallic surface of the critical-radius cylinder, and partly because the scattered polarization tends to become random as the trail ages and breaks up into blobs or glints.

When  $\lambda$  is replaced by  $\lambda \sec \phi$ , we obtain the forward-scatter echo duration

$$T_{ov} \simeq \frac{qr_e \lambda^2 \sec^2 \phi}{4\pi^2 D} = 7 \times 10^{-17} \frac{q \lambda^2 \sec^2 \phi}{D} \quad \text{sec} \quad (9-8)$$

We may carry over the philosophy, expressed in Sec. 8-10, that an acceptable definition of the transitional line density is given by that value of  $q$  which equates the underdense and overdense echo powers. From Eqs. (9-3) and (9-7) we obtain  $q_{tr} = 0.75 \times 10^{14}$  electrons/m, which is identical to the backscatter case (Hines and Forsyth, 1957).

Manning (1959a) has pointed out that these simple developments have not included the refractive effects of the electrons outside the critical radius. He has tackled this recalcitrant problem with the aid of a ray-tracing technique which was an extension of his earlier treatment of the backscatter trail, mentioned in Sec. 8-10. Manning finds marked deviations from the elementary forward-scattering cone of the underdense trail. After a few seconds, though, the overdense trail can no longer be considered an ideal cylinder. The blob-glint theory shows that the scattering pattern must broaden rapidly. The observational evidence (see Fig. 9-3) demonstrates that echoes with durations of 20 sec or longer become substantially omnidirectional.

An important feature of Manning's analysis is the deduction that the forward echo duration from even the ideal simple model should no longer bear the constant ratio  $\sec^2 \phi$  to the backscatter duration which Eq. (9-8) implies it should hold. Depending on the orientation of the trail Manning finds that a  $\sec^m \phi$  law may be adopted, with  $m \simeq 0.3$  for  $\beta = 90^\circ$ , up to

a maximum of  $m = 2$  for  $\beta = 0^\circ$ . The conclusions are assumed to apply from the moment of formation of the trail.

An experimental investigation of the dependence of the exponent  $m$  in  $\sec^m \phi$  was undertaken by means of simultaneous backscatter and forward-scatter observations (McKinley and McNamara, 1956). Pulses of 300-kw peak power were transmitted on 33 Mc/sec at Ottawa, and

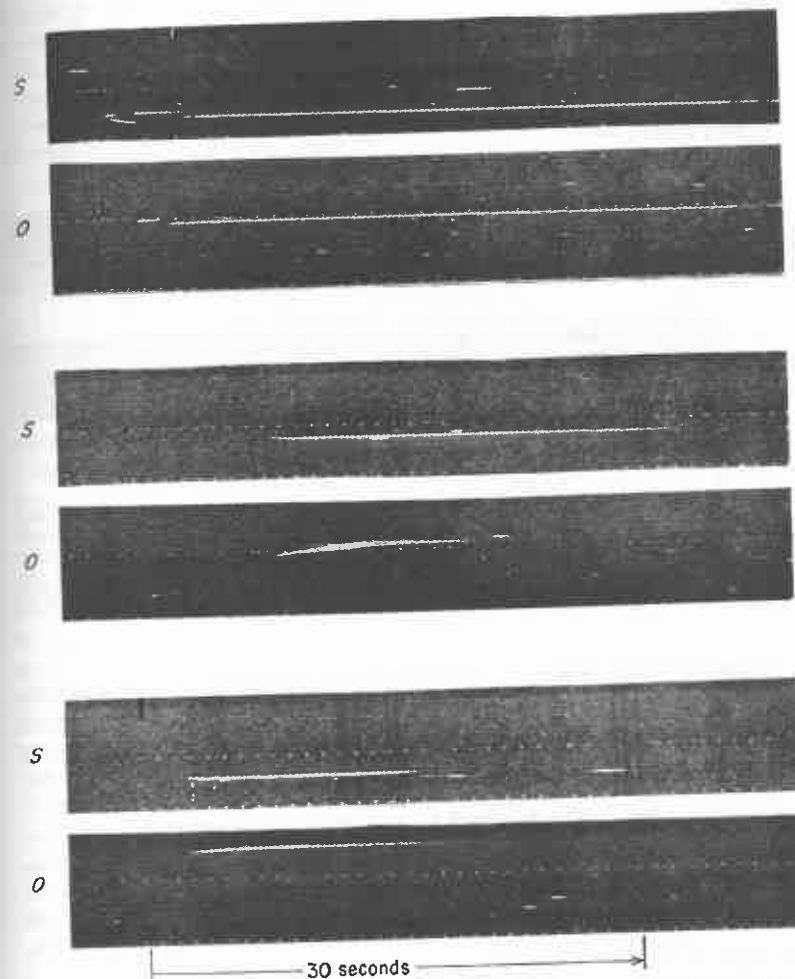


FIG. 9-3. Simultaneous records of meteor echoes obtained by backscatter at Ottawa (O), and by forward-scatter at Scarboro (S), 338 km distant from Ottawa.

meteoric echoes were recorded simultaneously on identical receiving equipments at Ottawa and at Scarboro (near Toronto), 338 km distant. Figure 9-3 shows some of the two-station records. Relatively few underdense trail echoes were simultaneously observed in comparison with the total number of such echoes seen at either station, which is a natural consequence of the condition that the underdense trail should be tangent both to a confocal ellipsoid about the two stations and to a sphere about the Ottawa station. A large fraction of the overdense echoes showed coincidences, the percentage increasing with the durations of the echoes. The Stanford group has confirmed this observation in an independent experiment (Villard, Peterson, Manning, and Eshleman, 1956). In the Ottawa-Scarboro test, most of the long-duration echoes appeared on both records

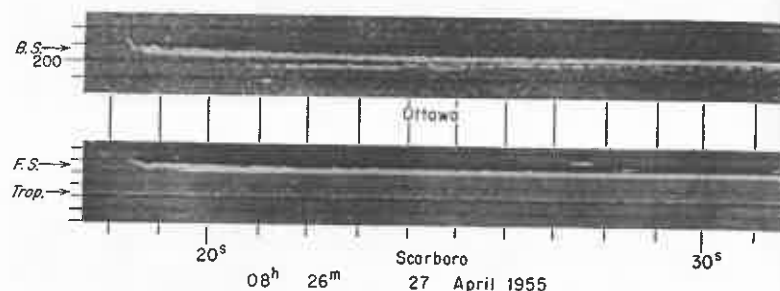


FIG. 9-4. A meteor echo observed simultaneously at Ottawa by backscatter (B.S.) and at Scarboro by forward-scatter (F.S.). The echo marked *Trop.* is the direct tropospheric wave between the Ottawa transmitter and the Scarboro receiver. The measured delay between the meteor echo and the direct wave, together with the Ottawa radar range, fixes the position of the meteor on the curve of intersection of an ellipsoid and a sphere.

—often with little or no delay after the appearance of the head echo associated with the passage of the meteor (see Fig. 9-4). From the geometry it was possible to compute  $\sec \phi$  with reasonable accuracy for every meteor. The exponent  $m$  was found from the ratio of the forward-scatter and backscatter duration. The underdense-echo class yielded the value  $m = 1.73$ , which was considered to be a good confirmation of the theoretical value  $m = 2$ , in view of the probable contamination of this group by an appreciable number of slightly overdense echoes. A value of  $m = 1.13$  was obtained for the overdense-echo group as a whole, which comprised all echoes with backscatter duration in excess of 3.5 sec. The group with durations longer than 15 sec gave  $m = 1.01$ .

In discussing these results we may note that the blob-glint theory, when applied to overdense meteor echoes, does not necessarily demand that  $m$

should be less than 2 at the beginning of the echo, but it does require  $m$  to approach unity as the trail ages (Eshleman, 1955). The theory proposed here by Manning (1959a) stresses that the value of  $m$  is mainly a function of the trail orientation  $\beta$ , while this is not a predominant factor in the blob-glint hypothesis. The Ottawa-Scarboro experiment was not designed to measure  $\beta$  for every meteor. If this were done, say, by means of a backscatter radar at the mid-point of the path, it should be possible to determine if there is a significant variation of  $m$  with  $\beta$  (McNamara and McKinley, 1959). An examination of the individual echo records of the original experiment does suggest that the  $\beta$  parameter may be important in some and not in others. Until further work is done we cannot decide the question. In the meantime, whatever may be the reasons, we do have the indisputable observational evidence that the duration enhancement of forward-scatter echoes from overdense trails is less than the factor  $\sec^2 \phi$  which would be predicted from the elementary model.

**9-3. Principles of intermittent communication via meteor trails.** For purposes of this discussion we shall divide the meteor population into three loosely defined groups. The first comprises the myriad of very faint meteors, of tenth to fifteenth magnitude and perhaps fainter, that are detectable near the noise level of a moderately powerful forward-scatter system. These echoes, which overlap in time to form practically a continuum, play a very important role in high-power ionospheric-scatter communication. This function will be examined in more detail in Sec. 9-7. The individual echoes in this group are too weak to support the intermittent type of low-power communication we are about to consider now. At the other extreme are the meteors of fifth magnitude and brighter which create overdense trails. These meteors are relatively few in number—typical rates may run from 10 to 100  $\text{hr}^{-1}$ —but their long durations tend to counterbalance their low rates. The effect of the long-duration echoes on meteor communication is usually beneficial though sometimes not. It is the intermediate group of meteors, beginning around the tenth magnitude and extending somewhat into the visual meteor range, which plays the major role in low-power meteor systems.

As was pointed out earlier, in order to support reflection of a signal between points A and B, an underdense trail should be tangent to one of a family of prolate spheroids having A and B as foci. The hourly rates of suitably oriented trails are dependent on the geographical disposition of A and B, on the system sensitivity, and on the distribution of meteor radiants and their activity. The useful integrated duration of such trails is usually a small fraction of the total time: typical values of this fraction, or duty cycle, run from 0.01 to 0.1. Means must therefore be devised first to determine when suitable trails are available

and second to transmit intelligence only during the usable durations of those trails.

The first of these aims is achieved by the closed-loop technique. A transmitter  $T_A$  and a receiver  $R_A$  are located at point A. Similar apparatus,  $T_B$  and  $R_B$ , are placed at point B.  $T_A$  and  $R_B$  operate on one frequency, and  $T_B$  and  $R_A$  are tuned to another frequency which differs enough from the first to permit simultaneous transmission and reception at each end of the link. So far, this arrangement does not differ from the usual point-to-point system where a direct line-of-sight path or a conventional ionospheric path is available. To find out when a suitable trail comes into existence,  $T_A$  continuously transmits an unmodulated carrier (for identification the carrier may be modulated by a tone or a code group). Reception of this carrier by  $R_B$  indicates that a suitable path has been established. The output of  $R_B$  turns on  $T_B$  and the wave from  $T_B$  travels to point A via the same meteor trail. The loop is now closed; both stations have learned that a suitable path exists, and either or both may proceed to send messages until the loop is broken by fading of the signal. The second object, to transmit intelligence only while a path exists, may be accomplished by storing the information temporarily, on magnetic tape, for example. When the loop is closed, the tape is started and run at high speed while the signal strength is above a predetermined level. The tape is then shut off to await the next closing of the loop. The transmission of a complete message may require many meteor trails occurring at random intervals.

An aptly descriptive name for the closed-loop type of low-power intermittent meteor communication was proposed by scientists of the Defence Research Board, at Ottawa, who demonstrated the first system in June, 1953 (Forsyth, Vogan, Hansen, and Hines, 1957). They called it JANET, after Janus, the Roman god of the doorway who looked both ways at once. The Stanford group, whose earlier work in forward-scatter-meteor propagation (one-way) was given full credit for inspiring the Ottawa experiments, have leaned toward the term "meteor-burst" system. The closed-loop and low-power features are not implicit in this description—open-loop ionospheric-scatter systems also depend to a great extent on the meteor bursts of countless small meteors. However, as a generic term "meteor burst" has the virtues of being expressible in lower-case letters and of being fairly readily understood by nonspecialists. Incidentally, in this connection the word "burst" applies to the sudden increase in the diffraction signal as the meteor crosses  $t_0$  and not to any physical bursting of the meteor along its path. The visual and photographic observers have always used "burst" in the latter sense, meaning a sudden flare due to disintegration of the meteor. It should be unnecessary to add that the "burst" reflection

is part and parcel of the whole diffraction echo and not a separate entity as some of the literature seems to suggest.

Naturally, it would be highly desirable to be able to predict the system performance for any time of day or year and for any physical disposition of the stations. We might start by attempting to derive a forward-scatter probability function similar to the  $P(Z)$  function discussed in Sec. 5-3. In backscatter geometry,  $\star$  meteors from a radiant at a zenith angle  $Z$  will be detected at an elevation angle  $Z$ , on a bearing  $180^\circ$  from the azimuth of the radiant. A plan position of density distribution of numbers of backscatter echoes, due to a uniform radiant distribution, may therefore be obtained directly from Fig. 5-6 by projecting the  $P(Z)$  values (in discrete steps for convenience in plotting) on a horizontal plane 100 km above the station (see Fig. 9-5a). A similar plot, with some radial displacement, will be obtained for the density of echo durations. A nonuniform radiant distribution will alter this picture. A concentration of radiants in the south, for example, will produce a concentration of meteor echoes in the north, as in Fig. 9-5b, but because of the spherical symmetry (omnidirectional antenna) the observed radar rates will still depend only on the zenith angles of the radiants. This statement will no longer be true if we separate transmitter and receiver, even if each station continues to use an omnidirectional antenna.

Figure 9-6a is a similar qualitative representation of the number density of echoes detectable by forward-scatter, again assuming a uniform radiant distribution (Villard, Eshleman, Manning, and Peterson, 1955). The concentration near the stations is partly because the range

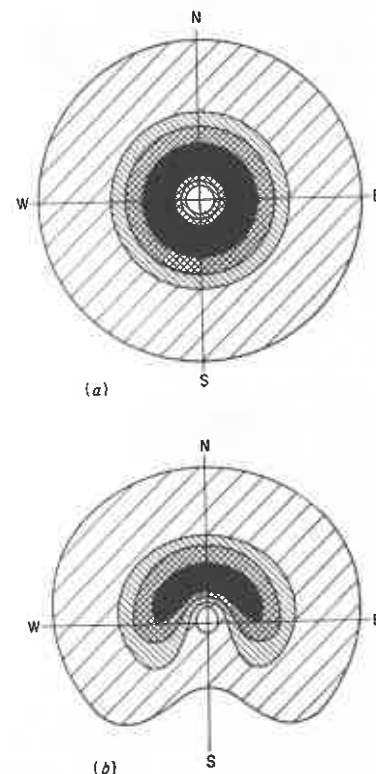


FIG. 9-5. (a) The relative number density of backscatter echoes, projected on a horizontal plane at a height of 100 km above the station; (b) the effect of a concentration of radiants south of the station's zenith.



attenuation [Eq. (9-7)] is less for these meteors than for trails near the mid-point. The orientation of the trail is also a factor which is evident particularly in the low-density region about the line between the two stations—the density per unit area of nearly horizontal trails is very low.

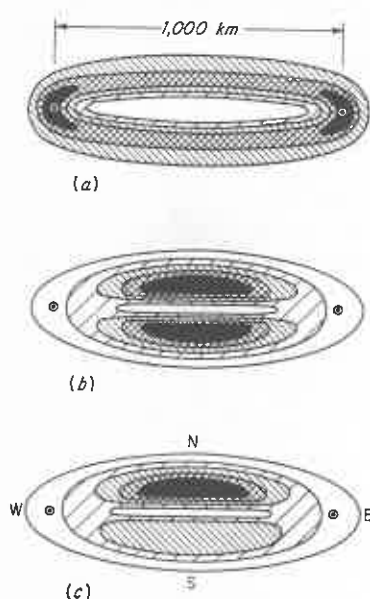


FIG. 9-6. (a) The relative number density of forward-scatter echoes, projected on a horizontal plane at a height of 100 km above the stations; (b) the relative duration density of forward-scatter echoes for a uniform radiant distribution; (c) the relative duration density for a radiant distribution concentrated in the south.

about the center line, though displaced toward the northern station. For optimum efficiency in a practical system it would therefore be advisable to use steerable antennas to take advantage of the shifting hot spots (Eshleman and Mlodnosky, 1957).

It is clear that we are not likely to be able to specify a simple probability function for the general forward-scatter situation. Given the specific geographical locations of the stations, the system parameters, and the radiant distribution, one can compute the performance. Until the radiant distribution of meteors as faint as the tenth magnitude becomes known for any time of the day or year, the pragmatic approach seems

We may now take each meteor represented in Fig. 9-6a and multiply its backscatter echo duration by  $\sec^2 \phi$  to obtain the plot of echo-duration density shown in Fig. 9-6b. The influence of the  $\sec^2 \phi$  factor is very marked; the maximum-density areas have shifted from near the stations to two "hot spots" located halfway between the stations, on either side of the mid-point. Obviously, to take full advantage of this one should use antennas of moderately high directivity and aim them at one or the other of the hot spots (or at both by means of split-beam antennas) rather than focus them on the mid-point where there are few suitably oriented meteors.

The effect of a concentration of radiants in the south is indicated in Fig. 9-6c. This situation is characteristic of an east-west path in the Northern Hemisphere around 0600 local time. For a north-south path, at the same time of day, the duration-density areas would be symmetrically disposed

best. That is, several experimental links are set up, with various spacings and orientations, and the observational data accumulated over a year or more (Forsyth and Vogan, 1955; Montgomery and Sugar, 1957; Meeks and James, 1957; Vincent, Wolfram, Sifford, Jaye, and Peterson, 1957b). From such data a reasonably good estimate can be made of the performance to be expected in a given practical situation. It is not a simple process to deduce the radiant distribution from the observed rates, nor does it usually lead to unambiguous results acceptable to the meteor astronomer.

#### 9-4. Information capacity of forward-scatter communication systems.

There are a number of factors to be considered in connection with the forward-scatter technique which have not been discussed in the earlier chapters because their effects on backscatter observations have not been very important. The theoretical and observational relations between integrated echo power, integrated echo duration, and radio wavelength are of great practical interest in forward-scatter. Absorption in the ionosphere and other environmental phenomena which are scarcely noticeable in backscatter work can influence forward-scatter propagation most markedly. In designing practical communication systems, the engineer's ingenuity must be brought to bear to devise suitable compromises between bandwidth, type of modulation, trigger level, and a variety of other parameters in order to cope with the unpredictable and highly variable characteristics of the meteor signal.

It has been shown that the hourly rate of backscatter echoes is proportional to the square root of the power sensitivity of the system (McKinley, 1951c). If we select an arbitrary minimum value of  $q$  as a reference level, the number of trails per hour with density  $q$  and higher is therefore

$$N \sim \frac{P_T^{1/2} G \lambda^{3/2}}{P_R^{1/2}} \quad (9-9)$$

The integrated duration of these echoes is  $\Sigma N_i T_i$ , where  $N_i$  is the number of echoes with duration  $T_i$ . In practice, the integrated backscatter duration can be evaluated directly from the radar range-time records, since the range resolution permits the analyst to make due allowance for overlapping of echoes in time whenever such overlapping becomes significant. We now define the *duty cycle*  $F$  as the fraction of the total observing time during which at least one echo is visible above an arbitrary power level; that is,  $F = (\Sigma N_i T_i) / T$  where  $T$  is the observing period. Where overlapping of echoes is appreciable and has not been allowed for in the reduction process, the true duty cycle may be taken to be approximately equal to  $1 - \exp(-F)$ . The uncorrected  $F$  values can exceed unity, but the true value can never quite reach unity.

Let us vary the wavelength while keeping the other parameters con-

stant in Eq. (9-9). If we assume  $T \sim \lambda^2$ , we have

$$F \sim N\lambda^2 \sim \lambda^{3.6} \quad (9-10)$$

From backscatter experiments the exponent of  $\lambda$  was shown to lie between 3.36 and 3.64 (McKinley, 1954b), which is reasonable confirmation of the theoretical value in Eq. (9-10).

If we allow one or more of the other system parameters to vary also, we may write

$$F \sim \frac{P_T^{1/2} G}{P_R^{1/2}} \lambda^{3/2} \quad (9-11)$$

This means that the system sensitivity should vary inversely as  $\lambda^7$  if we wish to keep the duty cycle constant. If  $P_T$  is a variable, it follows that  $P_T \sim \lambda^{-7}$ . If  $P_R$  is taken as the variable it is implied that the fraction of the total time during which echoes are present above a given power level should be directly proportional to  $\lambda^7$  (Villard, Eshleman, Manning, and Peterson, 1955).

Observations have shown that the exponent of  $P_T/P_R$  in Eq. (9-11) should be somewhat greater than 0.5. For convenience, let us tentatively label this exponent  $k$ . Backscatter data suggest as a minimum value  $k = 0.56$  (McKinley, 1954b), whence  $P_R \sim \lambda^{3.6/0.56} = \lambda^{6.3}$ . Forward-scatter experiments have yielded still higher values, ranging from  $k = 0.57$  to 0.96 (Forsyth and Vogan, 1955; Campbell and Hines, 1957). From these data the power dependence is found to vary between  $\lambda^{6.2}$  and  $\lambda^{6.6}$ . The decrease in the exponent of  $\lambda$  may be attributed to the influence of overdense trails. This has been borne out further by recent work done by J. C. Blair of the National Bureau of Standards. He found that  $\lambda^5$  represented the performance of a high-power scatter system during periods when meteor activity was high, compared to  $\lambda^8$  when few strong meteor signals were present. In theory, if we adopt  $T \sim \lambda$  for very-long-duration echoes (Sec. 8-10), we have  $F \sim \lambda^{3/2}$ , whence  $P_R \sim \lambda^5$ .

In a practical circuit one attempts to operate at a threshold signal level which bears a fixed ratio to the ambient noise level. This noise level is usually determined by cosmic noise, which has been found to vary about as  $\lambda^{2.3}$  (Sec. 2-8). If the receiver bandwidth is  $B$ , the threshold power level will then be proportional to  $B\lambda^{2.3}$ , which quantity replaces  $P_R$  in Eqs. (9-9) and (9-11). Let us also write the empirical  $k$  in place of the exponent of  $P_T$  and  $P_R$  in these equations. The duty cycle under these conditions becomes

$$F \sim \left( \frac{P_T G^2}{B \lambda^{2.3}} \right)^k \lambda^{3.5} \quad (9-12)$$

The average rate  $W$  at which information is transferred is proportional to the duty cycle multiplied by the bandwidth. Thus,

$$W \sim FB \sim P_T^k G^{2k} B^{(1-k)} \lambda^{(3.5-2.3k)} \quad (9-13)$$

Since  $k < 1$ , it follows that the information capacity of the system is increased by making  $B$  large, despite the decrease in duty cycle which necessarily occurs at the same time. A practical restraint on increasing the bandwidth indefinitely is determined by the length of time one is willing to wait between transmissions. Forsyth has suggested that one might vary the bandwidth and the signaling rate in proportion to the power received from each individual trail to make the most efficient use of the meteoric medium.

The diffraction-echo fluctuations due to the moving meteor do not cause any trouble because they have usually died out by the time the loop is closed and signaling started. Theoretically, all systems of modulation should be equally effective in meteor communications, though the current state of the engineering art may give one method a temporary ascendancy

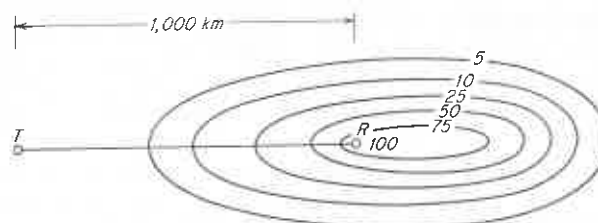


FIG. 9-7. Relative levels of information which might be intercepted by an independent receiver in the vicinity of the meteor scatter system.

over another. Frequency or phase modulation might be thought the most suitable in view of the rapid amplitude variation of the meteor signals. However, amplitude modulation has been used very successfully, either double-side band (Davis, Gladys, Lang, Luke, and Taylor, 1957) or single-side band (Vincent, Wolfram, Sifford, Jaye, and Peterson, 1957a). The bandwidths for teletype transmission extend up to 3 kc with these systems, and the signaling rates run up to 2,700 wpm or more. Transmitter output powers are usually of the order of 1 kw and antenna gains perhaps 10 db. The maximum distance between stations is limited to about 2,200 km, both by the earth's curvature and by the amount of antenna gain that is feasible at low elevations (Vincent, Wolfram, Sifford, Jaye, and Peterson, 1957b).

A certain amount of privacy is obtained in meteor communications by virtue of the directional characteristics of the reflections from underdense trails. Contours are shown in Fig. 9-7 of the relative level of information which could be intercepted by an unauthorized receiver located in the



vicinity of one of the stations (Forsyth, Vogan, Hansen, and Hines, 1957). Some protection from jamming transmitters or other interfering sources is also afforded. The presence of the overdense trails tends to spoil this privacy, because their scattering patterns are very broad. While few in number, compared to the underdense trails, their contribution to the integrated echo duration can be as much or more than that of the underdense trails. If the system loop is deliberately opened one second, say, after a meteor signal has closed it, regardless of the strength or the eventual duration of that signal, one may reduce this omnidirectional scatter at the expense of a substantially decreased duty cycle.

**9-5. Environmental effects.** During the period 1952 to 1956, when the forward-scatter systems were first being tested, it was found that the onset of an SID (sudden ionospheric disturbance) had little or no effect on the meteoric signal. The SID is a daytime absorption phenomenon caused by a sudden increase of ionization in the upper atmosphere, which in turn is due to abnormal ultraviolet radiation from a solar flare. World-wide h-f communications are sometimes interrupted for periods of a few minutes to a few hours when an SID occurs. The meteor-scatter links operate on frequencies in the vhf band well above the maximum usable frequencies for ordinary h-f propagation and are much less vulnerable to SID effects.

Besides its other attractive features, the comparative immunity of the JANET-type system to ordinary SIDs greatly intrigued the potential military and commercial users, particularly for use in high latitudes where conventional h-f circuits are more susceptible to trouble. However, it happened that all of the tests in this developmental period had been carried out south of the auroral zone. Furthermore, we were in the trough of the sunspot cycle at the time, and auroral activity was low. Some of the rosy optimism about the Far North application was rather rudely dampened when, in 1958, the peak of the sunspot cycle brought greatly enhanced auroral activity and severe absorptive effects.

In early July, 1958, a JANET 40-Mc/sec circuit was set up between Edmonton, Alberta, and Yellowknife, in the Northwest Territories, specifically to check out the performance in the auroral zone (Crysdale, 1960). Three days after it went into operation an intense polar-cap blackout on July 7 rendered it unusable for several hours. A polar-cap blackout, so called because its effects on radio communication are greatest from the auroral zones to the poles, is caused by a really gigantic flare on the sun—a 3<sup>+</sup> flare in the solar astronomer's terminology. As if just to emphasize the point, three more 3<sup>+</sup> flares occurred during August, 1958. In each case the ensuing hours were periods of complete radio silence in the auroral and polar regions, not only for the JANET system but also for high-power ionospheric-scatter equipments and conventional h-f stations.

This was a pretty rugged environmental test to be imposed on the new system in the first few days of its northern trials, particularly as it is estimated that not more than seven or eight 3<sup>+</sup> flares occurred in all of 1958, a year of most intense solar activity! ★

The initial reaction of disappointment was followed by the realization that this early encounter with the worst conditions that are ever likely to occur may have avoided a good deal of trouble later. The prospective customers, who wanted to use the system in the north, naturally adopted a wait-and-see attitude, while the designers hastened back to their drawing boards.

The polar-cap blackout can be overcome by a shift to higher frequencies. To see this, let us look briefly at the nature of the polar-blackout absorption. One of the best instruments for evaluating the ionospheric absorption of vhf waves is the riometer. This is nothing more than a sensitive receiver, operating on a fixed frequency  $f_A$  (30 Mc/sec is a commonly used frequency), which records cosmic-noise signals from a broad cone overhead. During the most severe polar blackout the attenuation of cosmic noise on its one-way, nearly vertical path through the ionosphere may exceed 10 db, though it very rarely reaches 15 db. We assume that the major part of this attenuation occurs below the meteoric region (some recent evidence suggests that it may be as low as 50 km at times). We may then calculate the absorption loss incurred along any oblique path passing twice through this region, from the formula

$$\text{Loss in db} \simeq 2A_V \left( \frac{f_A}{f} \right)^2 \sec \phi \quad (9-14)$$

where  $A_V$  is the riometer value of the vertical absorption in decibels and  $f$  is the frequency used on the oblique path. Equation (9-14) shows that when  $A_V$  reaches 10 db on 30 Mc/sec, as it can easily do after a 3<sup>+</sup> flare, the absorption attenuation on a 40-Mc/sec circuit between two stations 1,000 km apart ( $\sec \phi \simeq 4.5$ ) will amount to 50 db. Small wonder that the 40-Mc/sec link was put out of operation! On the other hand, a similar circuit working on 100 Mc/sec should suffer only 8 db loss during the same blackout conditions (Crysdale, 1960). Tests are now in progress on both 40 Mc/sec and 100 Mc/sec between Goose Bay, Newfoundland, and Ottawa, to find out if this prediction will be verified experimentally. The duty cycle falls off rapidly with increasing frequency, and it remains to be seen if practical low-power systems can be built to work at frequencies of 100 Mc/sec and higher. Recent high-power experiments on 200 Mc/sec over a 1,300-km path have shown the possibilities to be encouraging (Heritage, Weisbrod, and Fay, 1960). The individual echo durations were naturally quite short, but the duty cycle was as high as 20 per cent at times.

A certain amount of absorption may also accompany severe auroral activity, and though it can be almost as strong as the polar-cap-blackout absorption, it is not nearly as sustained. Considerably more important than the absorption effects are the multipath signals which are scattered from the ionized regions associated with the aurora. The extremely rapid power fluctuations caused by this nonmeteoric type of propagation will introduce serious message errors in a system designed to suit meteor signals only. Increasing the frequency does not help because the frequency dependence is roughly the same for both—after all, auroral targets are analogous to meteors in that they are presumed to consist of lines of ionization created by high-speed particles ejected by the sun. Since we cannot easily ignore the contribution from the auroral mode of propagation, it behooves us to devise ways and means of utilizing it to advantage in meteor-scatter systems. Some steps toward this end are being taken: one of the most promising is the incorporation of sophisticated error-correcting techniques in the data-handling part of the system. As the duty cycle generally is greater during times when auroral signals are present, it seems likely that satisfactory information rates may still be obtained, even though a fair fraction of the time may be occupied by the redundancy involved in reading back and correcting errors.

There are several other modes of propagation which can occasionally intrude adversely. One of them appears to be backscatter from the ground at great distances, received via the ionosphere. For example, a ray from the transmitter may pass through the meteoric region and be refracted by a higher ionospheric layer (the  $F$  region) to the ground. The ground-reflected signal may follow another path via the  $F$  region to the receiver. The delays between the meteor signal and the ground-scatter signal can amount to 20 or 30 msec, and this unwanted signal can create errors in the transmissions. Error-correcting techniques may be of some help. A move to higher frequencies would also reduce the ground scatter. The relatively large delays suggest that a slow FM technique might be fairly effective against this type of interference. For example, if the transmitter frequency were varied in a saw-tooth fashion (or in discrete steps) where the period of the saw-tooth waveform was just longer than the delay of the ground scatter, the receiver (which would be tracking the transmitter frequency) could reject the off-frequency ground echoes.

From this brief discussion the reader will no doubt conclude that nature in her worst moods can be very devastating in this as in other fields. The applied scientist and the design engineer must work unceasingly to circumvent her, or if they cannot ward off the slings and arrows of outrageous fortune, they must seize them and turn them to their own advantage. The inevitable result seems to be that the practical embodiment of a beautifully simple original concept soon becomes hedged in by a fear-

some array of complex gadgetry. In this respect, though, the meteor-scatter system is no worse, if no better, off than any other modern mode of communication, and the evolutionary process often does much to advance man's knowledge in rather unexpected areas. Quite possibly the inventors and users of fire and smoke signals were forced into operational research on the frequency of incidence of fog and rain, thereby establishing the science of meteorology.

Even in high latitudes, though, these abnormal environmental effects are either infrequent or can be overcome. They are much less troublesome over the equatorial and temperate zones of the earth's surface. Meteor-scatter systems are now in production and are entering service in several countries because they offer features which complement rather than supplant the characteristics of other types of radio communications. Certain circumstances even favor meteor-scatter for air-to-ground communications, despite the intermittent nature of the contact and the complex ancillary equipment which must be carried.

**9-6. Meteor-scatter equipments.** Fully engineered meteor-scatter equipments have been produced by industrial firms both in Canada and the United States. The following brief description of some features of these systems is intended to cover only a few high lights for the information of the reader who may not be versed in communication techniques. The expert will find in the referenced papers much more detail than we can provide in this book—the highly specialized terminology of communications engineering would alone take several pages to explain to the uninitiated.

The JANET equipment, of which a number have been supplied to government agencies in Canada, England, Holland, and the United States, is a typical representative of the closed-loop systems (Davis, Gladys, Lang, Luke, and Taylor, 1957). It is designed to accept and discharge information continuously at the standard teletype rate of 60 wpm. Since the average duty cycle is about 5 per cent, transmission over the radio links must take place at twenty times this rate, or roughly 1,200 to 1,300 wpm, to maintain the steady flow. To do this, temporary storage must be provided for the incoming and outgoing information. The perforated paper tape accumulated in the input store is "read" electronically and converted to a suitable code for radio transmission. At the receiving end the snatches of coded messages are applied to high-speed magnetic tape. The tape is then read continuously, at a lower speed, into a teleprinter which produces typed text.

In the quiet periods when no meteor path is available, each JANET transmitter emits an unmodulated carrier on different frequencies,  $f_1$  and  $f_2$ , spaced about 1 to 2 Mc/sec apart in the band 40 to 50 Mc/sec. Figure 9-8 is a block diagram of one terminal of the JANET system; the other

terminal is identical except that frequencies  $f_1$  and  $f_2$  are reversed. The incoming carrier level in each receiver is compared with the ambient noise level in sidebands outside those required for intelligence. When the carrier levels rise to a predetermined value, the master gate controls are actuated at each end. These open both the input gates and the receiver gates to permit synchronizing signals from one station to pass to the other. After synchronization has been achieved the output gates are opened, and the transmission of messages begins in either or both directions. Transmission is stopped when the carrier level drops below threshold, the output store is filled, or a "stop-code" signal is received.

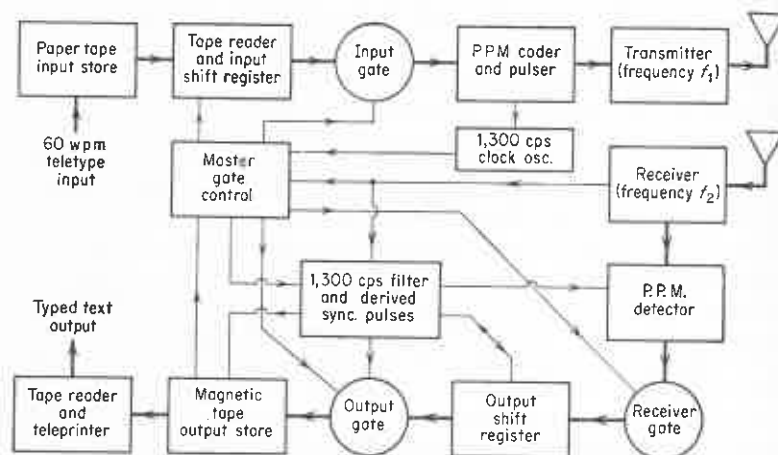


FIG. 9-8. Simplified block diagram of one terminal of a JANET system. Heavy lines show the flow of information; light lines indicate the control paths.

During a transmission, the teletype paper tape is removed from the input store by means of a fast-acting stop-and-start drive mechanism. The standard 5-digit teletype code is read into the input shift register, passed through the open input gate, and converted into a two-position PPM code (pulse position modulation). This PPM is applied to the carrier as amplitude-modulated pulses, which have a cosine-squared waveform to minimize unwanted higher sidebands. The 1,300-cps clock oscillator superposes a synchronizing tone on the PPM. At the receiver, this 1,300-cps tone is filtered out and subdivided to provide the necessary timing and synchronizing pulses.

The output of the PPM detector goes through the receiver gate into the output shift register. When the system is synchronized and the output gate is opened, the shift-register output is applied to the magnetic tape

on five parallel tracks. In effect, the PPM code has been converted back to teletype code at this stage. The accumulated tape is drawn continuously from the output store, put through a special tape reader, and finally printed out by a conventional teleprinter.

The five racks which house the data-handling portion of one terminal of the JANET 204-B system are shown in Fig. 9-9. The detail of the magnetic-tape rate-changing equipment appears on Fig. 9-10. This unit is the heart of the output store; it records the incoming information intermittently at high speed and reads it out continuously at low speed. Figure 9-11 is a view of the antennas.

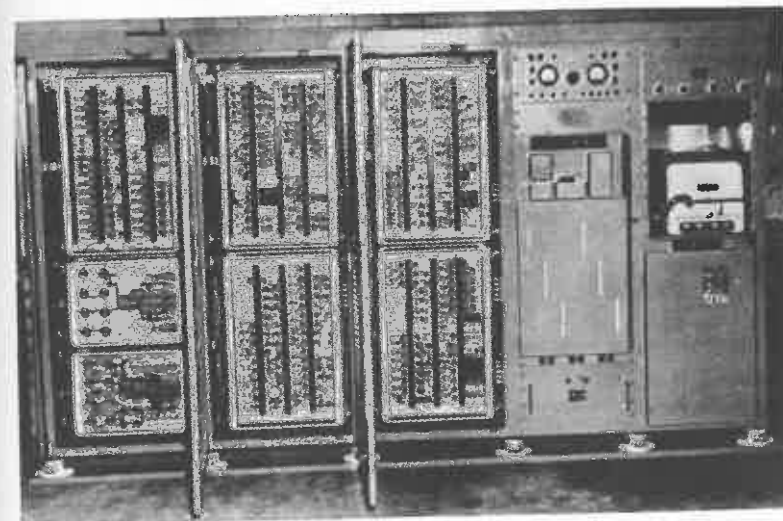


FIG. 9-9. Data-handling equipment of JANET 204-B system. (Courtesy of G. W. L. Davis, Ferranti-Packard Electric, Ltd., Toronto.)

In broad outline, the experimental equipment built by the National Bureau of Standards (Carpenter and Ochs, 1957) was similar to the JANET apparatus. It provided duplex teletype channels with 60 wpm input and output, and the high-speed part of the system operated at 2,400 wpm or more.

The meteor-scatter system designed by the Stanford Research Institute was also similar in operating principle to JANET, though there were many differences in detail (Vincent, Wolfram, Sifford, Jaye, and Peterson, 1957a). For example, over the radio link the intelligence was transmitted in a binary code by switching back and forth from one to the other of two closely spaced carrier frequencies. For convenience, a separate

radio link was used for control purposes in the early experimental setup. Two kinds of output storage were available, the high- and low-speed-magnetic-tape method, and a magnetic-core storage wherein the informa-

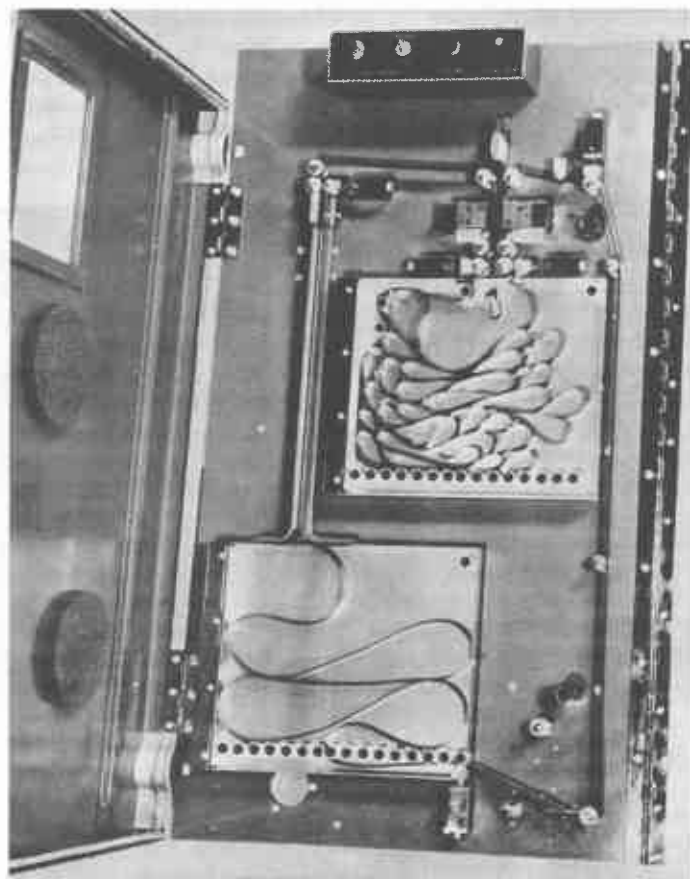


FIG. 9-10. Magnetic-tape rate-changing equipment. (Courtesy of G. W. L. Davis, Ferranti-Packard Electric, Ltd., Toronto.)

tion was stored in magnetic-core matrices and read out at slow speed into a standard teleprinter.

Voice transmission was also accomplished with the Stanford system. Figure 9-12 illustrates the one-way arrangement for doing this. Recording was done at normal speed on a continuous loop of magnetic tape.

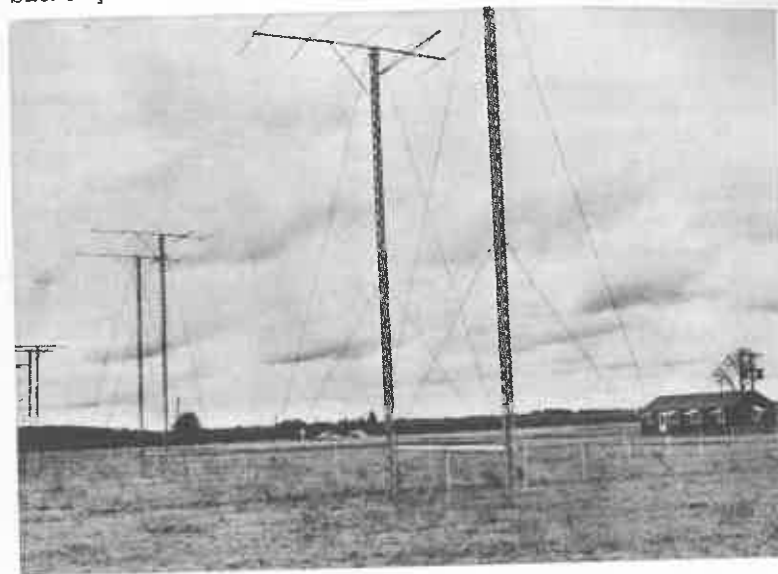


FIG. 9-11. JANET antenna system. (Courtesy of G. W. L. Davis, Ferranti-Packard Electric, Ltd., Toronto.)

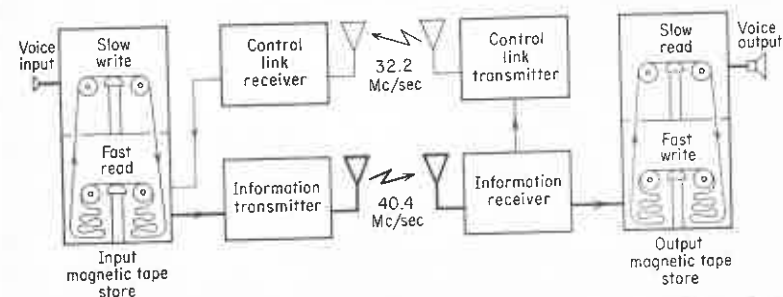


FIG. 9-12. The Stanford arrangement for transmitting speech via meteor trails. (Vincent, Wolfram, Sifford, Jaye, and Peterson, 1957a.)

The slack in the tape was stored in bins on either side of the high-speed read-out head. On command from the link control the voice signal was read out at five times normal speed and used to modulate a single-side band transmitter. The process was reversed at the receiving end. Good quality voice communication was obtained with 20-kc/sec bandwidth over the high-speed part of the circuit.

Facsimile pictures have been successfully transmitted over a 1,450-km

path via meteor trails (Bliss, Wagner, and Wickizer, 1957). Two frames per second were sent, each with a resolution of 67 picture elements per inch, which required a bandwidth of 27 kc. The early experimental arrangement did not use a closed loop; the pictures were repeated until one or more complete ones got through. The results were sufficiently encouraging to warrant further development.

An air-to-ground meteoric-scatter system has been developed as a cooperative project between the Wright Air Development Center and the Hughes Aircraft Company. This has some special features well worth mentioning (Hannum, Evans, Chambers, and Otten, 1960). In the first place it is not solely a meteor-scatter system. Other modes of propagation, such as auroral ionization and sporadic *E*, can be turned to advantage as well. This is partly because both links operate on the same frequency, which avoids the lack of correlation and reciprocity that sometimes occurs with two-frequency operation when modes other than the meteoric one appear. More important, perhaps, the error-correcting technique incorporated in the system ensures that garbled parts of the message will be checked and corrected or repeated until correctly received.

The ground transmitter probes the medium continuously with a short-pulse code while the aircraft transmitter is silent. On reception of the ground signal, via a meteor trail or other path, the airborne transmitter replies on the same frequency, either to send back a message or to invite the ground station to send its traffic. Time-sharing of groups of pulses from either end of the link permits single-frequency operation. This results in some delay, because it is not possible to transmit while receiving, but higher effective transmission rates (2,700 wpm) help to offset the delay.

The radio link in the Hughes system employs a type of frequency shift keying which may be better described as dual-channel amplitude modulation, something like the Stanford system. Both the airborne and ground transmitters have 10-kw peak pulse capacity (Fig. 9-13). It is scarcely practical to provide directivity in the airborne antenna, but the over-all loop gain can be maintained by using a high-gain ground antenna (Fig. 9-14) which should be kept pointed in the general direction of the aircraft.

**9-7. High-power ionospheric scatter.** In 1951 an historic experiment was performed to test the theory that scattering from ionospheric turbulence in the lower *E* region could support a weak but continuous signal at vhf frequencies over distances well beyond the line of sight (Bailey and seven others, 1952). A cw transmitter at Cedar Rapids, Iowa, radiated 25 kw on 49.8 Mc/sec. The receiver was located at Sterling, Virginia, 1,245 km away. From each terminal, antennas with gains of 18 db were directed to the mid-point of the path so that their patterns overlapped roughly at the 100-km level. The experiment was a success: a weak level

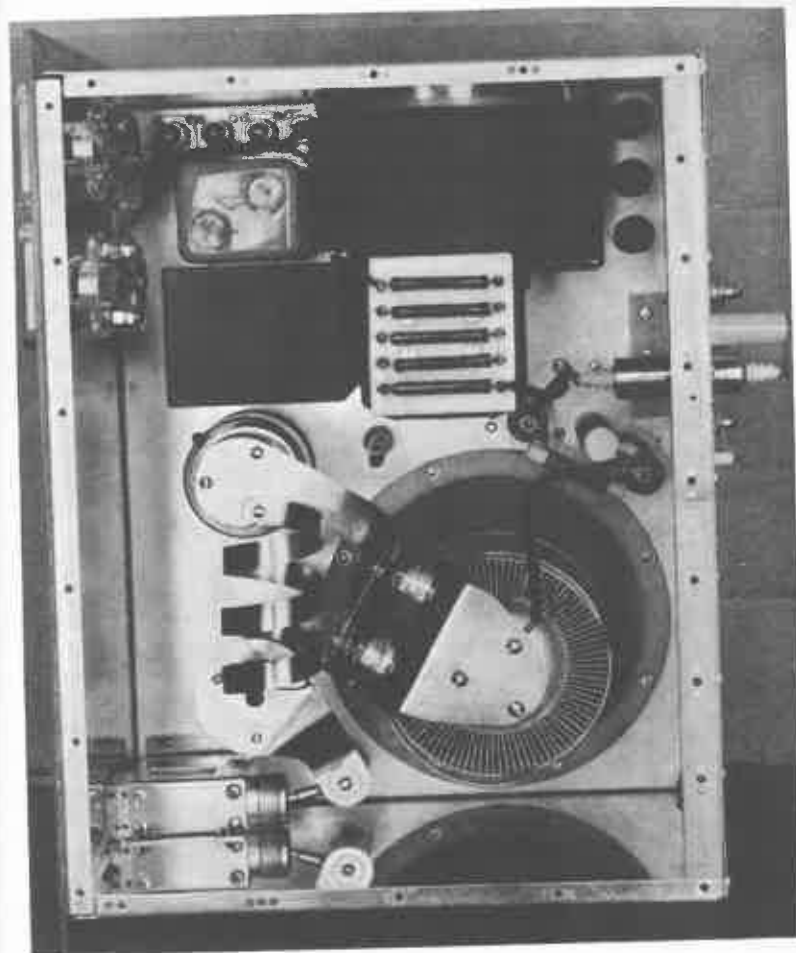


FIG. 9-13. Airborne 10-kw 50-Mc/sec power amplifier. (Courtesy of L. C. Parode, Hughes Aircraft Company, Los Angeles.)

of signal was maintained day in and day out. Superposed on the weak signal were numerous strong spikes which were obviously due to echoes from meteors. There was never any question about the meteoric contribution to the total signal level in those cases where the individual meteor echoes could be identified. In fact, the continued operation of this circuit did much to aid forward-scatter research on meteors, since the larger meteoric signals were detectable well off to either side of the path

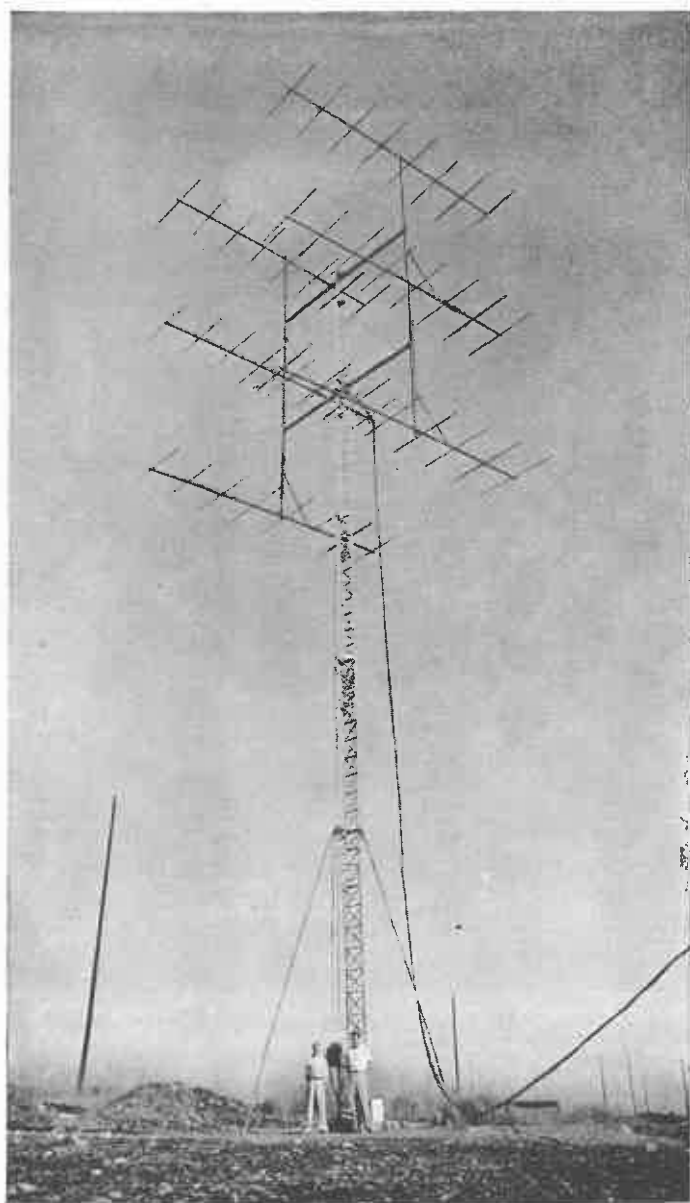


FIG. 9-14. Forty-eight-element ground antenna. (Courtesy of L. C. Parode, Hughes Aircraft Company, Los Angeles.)

between the stations. However, it was not so easy to resolve the relative contributions of meteoric and other forms of ionization to the continuous background signal level.

This weak signal showed diurnal variations that were similar in broad outline to the typical backscatter records of meteor echoes which were available at that time. There were discrepancies, though, such as a relatively higher daytime level and smaller swings between maxima and minima of the scatter signal, which were significant enough to suggest that small meteors could neither be the sole agency nor even an important one. This conclusion was given added weight by some early measurements of the heights from which the scattering occurred; an observed daytime level of 70 km appeared to be well below the normal meteor levels.

Later developments corrected some of these earlier impressions. The daytime scattering heights were revised upward to around 80 km through more accurate measurements. The lack of close agreement between the diurnal curves for the scatter signal and the backscatter meteor echoes was resolved by the Stanford and Ottawa workers who brought out a point which we discussed in Sec. 9-3, namely, that a given radiant distribution will yield forward-scatter daily-rate curves which depend strongly on the geometry and which can differ markedly from the usual curves obtained by backscatter radar. In particular, the forward-scatter meteor results showed higher midafternoon rates and a smaller ratio of maximum-to-minimum rates (compare curves *B* and *D*, Fig. 5-4). Other experiments demonstrated that the innumerable faint meteors could provide (not necessarily that they did provide) the continuum of scattering centers required to support the weak scatter signal (Villard, Eshleman, Manning, and Peterson, 1955; Forsyth and Vogan, 1955; McKinley, 1954b).

The antennas in the Cedar Rapids-Sterling tests were beamed directly above the mid-point of the path because scattering from ionospheric turbulence would favor this on-path propagation. On the other hand, meteoric propagation would tend to be off path, via either of the two hot spots of Fig. 9-6b. The gain of the antennas in the vicinity of these hot spots was appreciably lower than the gain at the mid-point, and in this way the original arrangement discriminated against the meteoric component. The relative contribution of each mode to the total signal might have been determined by steering the antennas rapidly from the center point to either side, but this was difficult to do. Instead, two additional receiving stations were set up on either side of Sterling with high-gain antennas beamed at the mid-point of the original path (Pineo, 1958). This meant that the central region illuminated by the transmitter became the southern hot spot for the receiver north of Sterling, and vice versa, while the Sterling receiver continued to provide on-path records for com-



parison. The results are shown in Fig. 9-15, where the signal gain of each of the off-path links is plotted relative to that of the on-path link. The crosshatched areas, during which the off-path gain was positive, show that for most of the day the meteor signal predominated.

The weight of the evidence seems to be that the bulk of the weak scatter signal is made up of the contribution of very faint meteors, perhaps as faint as fifteenth magnitude, but that during part of the daytime an additional agent is certainly present, probably photo-ionization coupled with increased turbulence, which becomes effective in the afternoon when

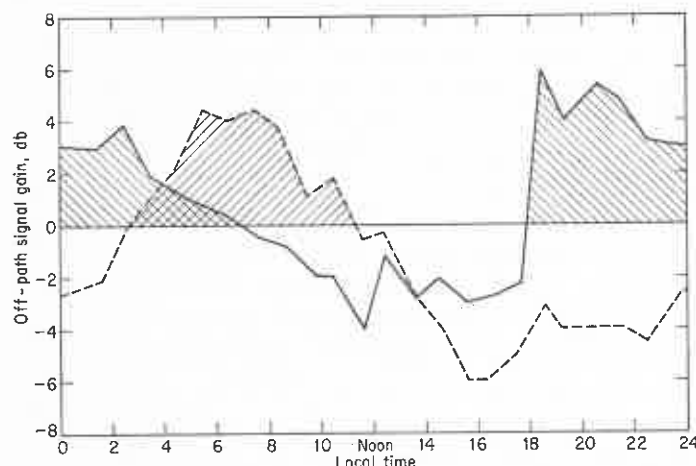


FIG. 9-15. The variation with time of the signal gain of the two off-path receivers relative to zero db for the on-path receiver. The solid curve refers to the path utilizing the southern hot spot and the dashed curve applies to the northern hot spot. The extent of the shaded area shows that the meteor signal predominated during all but the afternoon hours (after Manning and Eshleman, 1959, based on data by Pineo, 1958).

meteor activity is at a low ebb. The interesting possibility remains that meteor-trail ionization might be enhanced or prolonged by the solar radiation process too. As was pointed out in Sec. 8-12, photo-ionization may reduce the attachment coefficient and increase the duration of the meteor echo. This would apply to the longer echoes mainly, which tend to scatter omnidirectionally; hence the effect would be observed on path as much if not more than off path. Both photo-ionization and photodetachment are enhanced during a normal SID, and so either mechanism could account for the observed increase in the weak scatter signal during an SID.

If the increase in photo-ionization is great enough and occurs sufficiently low in the atmosphere, the signal level will decrease rather than rise, owing to absorption setting in. The catastrophic polar-cap blackout cuts off the high-power ionospheric-scatter circuits just as effectively as it severs the low-power closed-loop links. The obvious solution is the same as the one advocated for the meteor system, namely, to move to a much higher frequency, but it may be difficult to implement.